



INSTITUT TEKNOLOGI DEL

MATERI PRAKTIKUM

4233201 – Kecerdasan Buatan

Semester Ganjil Tahun Ajaran 2026/2027

Session Date	: 05 Oktober 2026
Week/Session	: 5/2
Topic	: K-Means
Practicum Goals	: Mahasiswa mampu melakukan klasterisasi menggunakan K-Means
Duration	: 2 x 50 minutes
Place of Submission	: ecourse
Submission Deadline	:

K-Means Clustering digunakan untuk mengelompokkan data ke dalam beberapa kluster berdasarkan kesamaan karakteristik. Analisis ini menggunakan dataset Mall Customers, yang memuat berbagai informasi pelanggan. Sebagai langkah awal, proses klusterisasi akan difokuskan pada dua fitur utama, yaitu Annual Income dan Spending Score, untuk menyederhanakan pemahaman serta mempermudah visualisasi kluster. Pendekatan ini memungkinkan interpretasi hasil yang lebih intuitif sebelum melibatkan fitur lainnya.

```
In [1]: import pandas as pd
df = pd.read_csv('/content/Mall_Customers.csv')
df.head()
```

```
Out[1]:
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
In [2]: # cek missing values
df.isnull().sum()
```

```
Out[2]:
```

	0
CustomerID	0
Genre	0
Age	0
Annual Income (k\$)	0
Spending Score (1-100)	0

dtype: int64

```
In [3]: df.describe()
```

```
Out[3]:
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

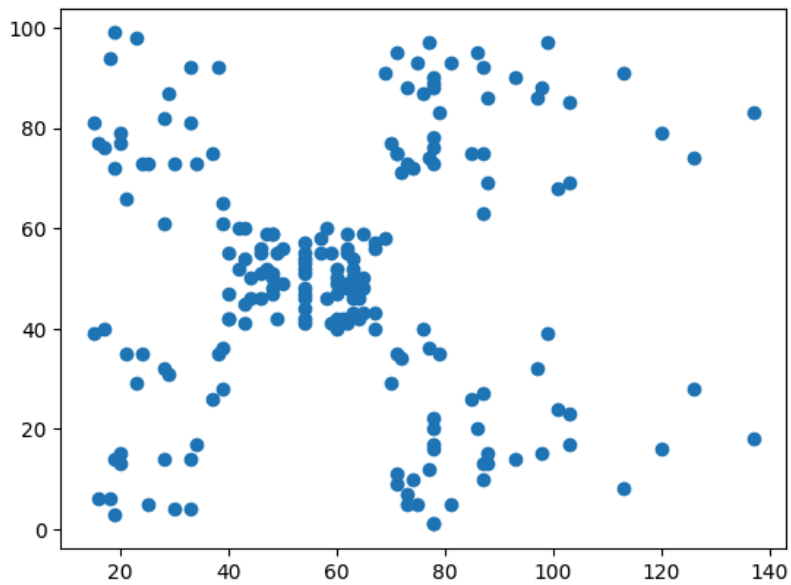
```
In [4]: dataset = df.iloc[:,3:]
dataset.head()
```

```
Out[4]:
```

	Annual Income (k\$)	Spending Score (1-100)
0	15	39
1	15	81
2	16	6
3	16	77
4	17	40

```
In [5]: x = df.iloc[:, 3]
y = df.iloc[:, 4]
```

```
In [6]: import matplotlib.pyplot as plt
plt.scatter(x, y)
plt.show()
```



```
In [7]: import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import MinMaxScaler

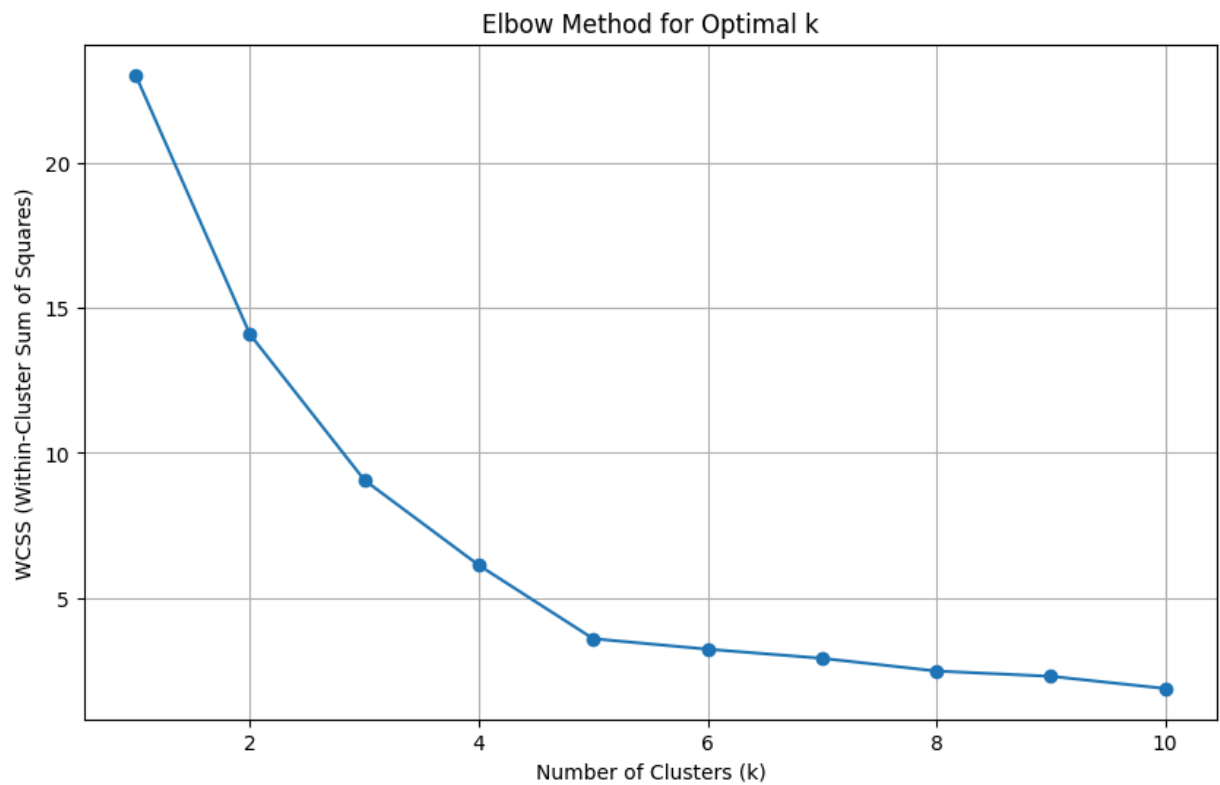
# Scaling data dengan MinMaxScaler
scaler = MinMaxScaler()

# Mengambil kolom X (index 3) dan Y (index 4) dari DataFrame
X = pd.DataFrame({'Annual Income (k$)': x, 'Spending Score (1-100)': y})

X_scaled = scaler.fit_transform(X)

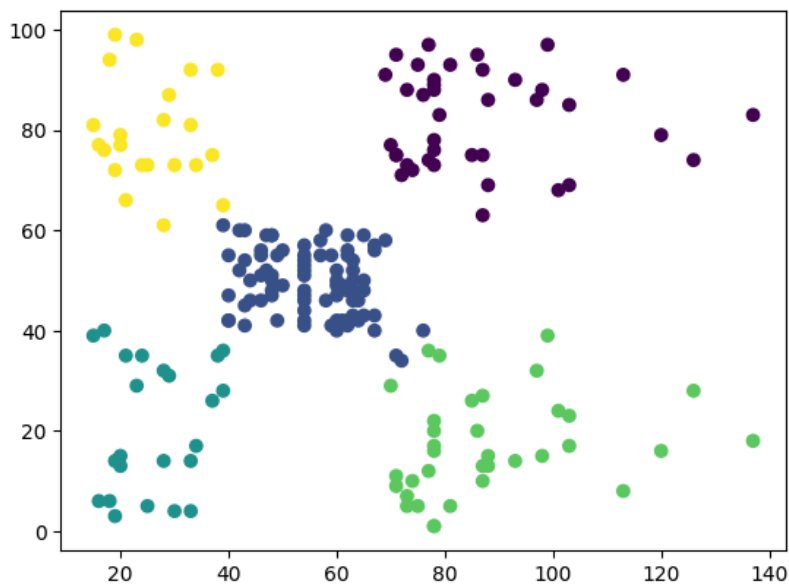
# Menentukan jumlah kluster optimal menggunakan Elbow Method
wcss = []
for i in range(1, 11): # Coba k dari 1 sampai 10
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Membuat plot Elbow Method
plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), wcss, marker='o', linestyle='--')
plt.title('Elbow Method for Optimal k')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('WCSS (Within-Cluster Sum of Squares)')
plt.grid(True)
plt.show()
```



```
In [8]: # dari elbow method, nilai k optimal untuk data ini adalah 5
kmeans = KMeans(n_clusters=5)
kmeans.fit(X)

# Plot hasil klaster
plt.scatter(x, y, c=kmeans.labels_)
plt.show()
```



Selanjutnya klasterisasi dilakukan dengan menggunakan tiga fitur utama, yaitu Age, Annual Income, dan Spending Score.

```
In [9]: from mpl_toolkits.mplot3d import Axes3D
import numpy as np

# Data Age, Annual Income, dan Spending Score
x = df.iloc[:, 2]
y = df.iloc[:, 3]
z = df.iloc[:, 4]

# Membuat plot 3D
fig = plt.figure(figsize=(13, 13))
ax = fig.add_subplot(111, projection='3d')

# Plot data
ax.scatter(x, y, z, c='red', marker='o')

# Label sumbu
```

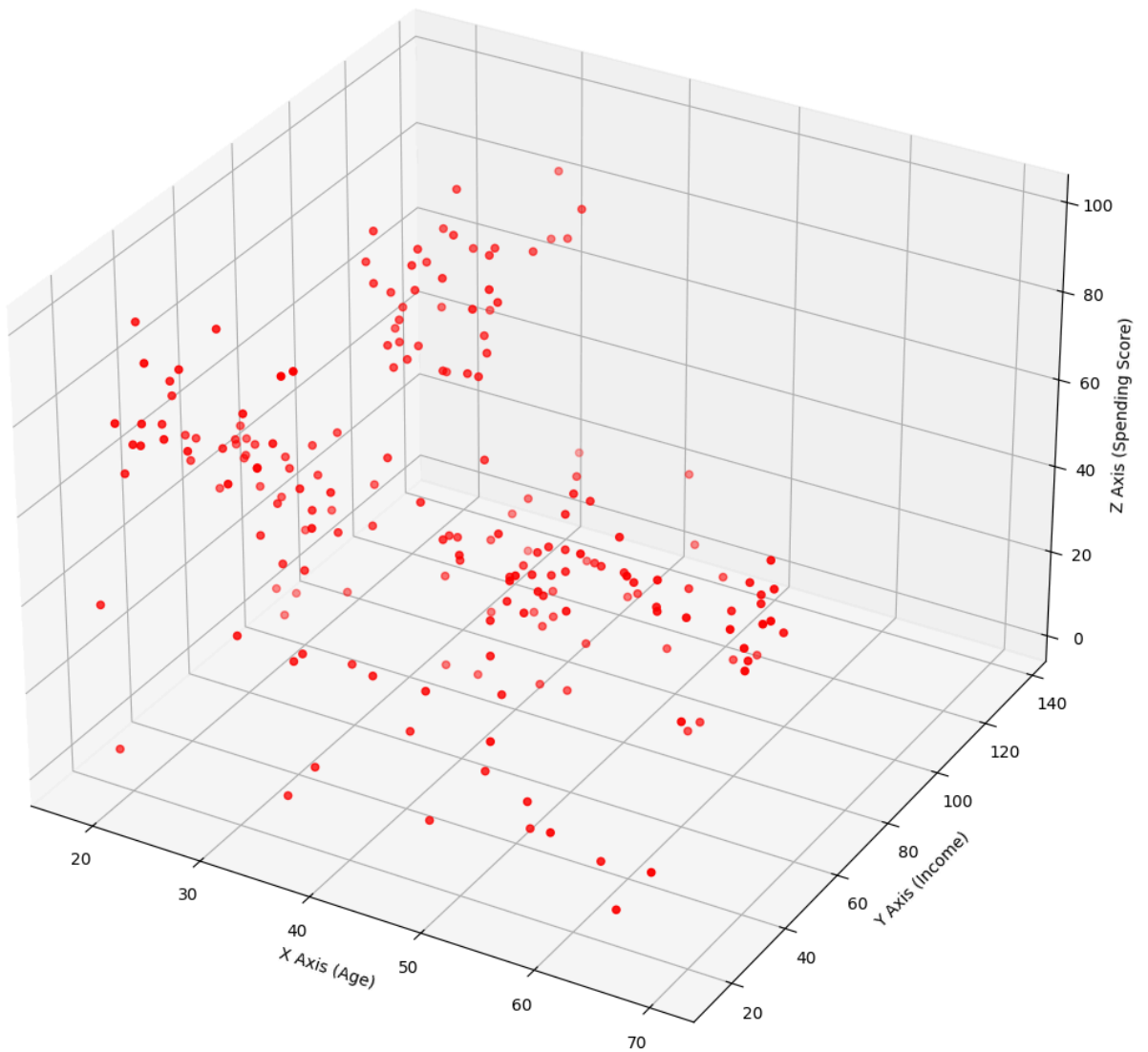
```

ax.set_xlabel('X Axis (Age)')
ax.set_ylabel('Y Axis (Income)')
ax.set_zlabel('Z Axis (Spending Score)')

plt.title('3D Scatter Plot with Matplotlib')
plt.show()

```

3D Scatter Plot with Matplotlib



```

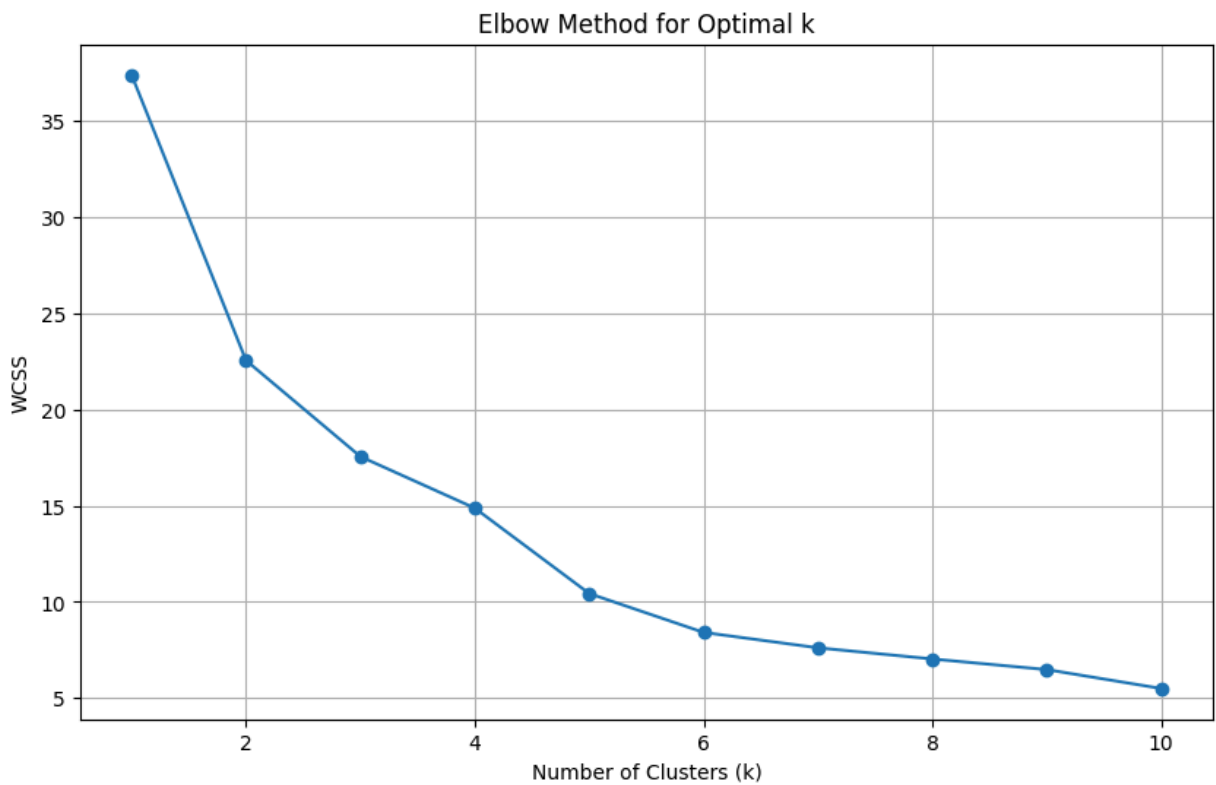
In [10]: # Menggabungkan data X, Y, Z menjadi array untuk clustering
X = np.array(list(zip(x, y, z)))

# Scaling data dengan MinMaxScaler
scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

# Menentukan jumlah kluster optimal dengan Elbow Method
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Visualisasi Elbow Method
plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), wcss, marker='o', linestyle='--')
plt.title('Elbow Method for Optimal k')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('WCSS')
plt.grid(True)
plt.show()

```



```
In [11]: # Pilih k optimal setelah melihat Elbow
kmeans = KMeans(n_clusters=6, random_state=42)
clusters = kmeans.fit_predict(X)

# Menambahkan hasil klaster ke DataFrame
df['Cluster'] = clusters

# Visualisasi 3D Hasil K-Means

# Plot data dengan warna berdasarkan klaster
fig = plt.figure(figsize=(13, 13))
ax = fig.add_subplot(111, projection='3d')

# Plot setiap titik dengan warna sesuai klaster
scatter = ax.scatter(x, y, z, c=clusters, cmap='viridis', marker='o')

# Label sumbu
ax.set_xlabel('Age')
ax.set_ylabel('Annual Income (k$)')
ax.set_zlabel('Spending Score (1-100)')

# Menampilkan Legenda klaster
legend1 = ax.legend(*scatter.legend_elements(), title="Clusters")
ax.add_artist(legend1)

plt.title('3D Scatter Plot with K-Means Clustering')
plt.show()
```

3D Scatter Plot with K-Means Clustering

